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CHEMICAL REACTION ENGINEERING (CHE 321)

QUIZ 2

CLOSED BOOK, TIME 20 MIN

Gas phase reaction $2A + B \rightarrow 2C$ takes place in an isothermal packed bed reactor with the following rate law:

$$r'_A = -k P_A^{1/3} P_B^{2/3}$$

The molar flow rates of the feed stream are 0.7, 0.7, and 1.4 mol/s for A, B, and an inert species.

Write two ordinary differential equations that must be solved simultaneously to determine the conversion of the limiting reactant and the pressure as functions of catalyst weight along the reactor.

Your answer may be in terms of the following constants:

$P_0 = 1 \text{ bar}$ is the inlet pressure and temperature

$k = 0.08 \frac{\text{mol}}{\text{s} \cdot \text{kg} \cdot \text{Pa}}$ is the rate constant at the reactor's temperature.

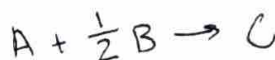
$\alpha = 0.02 \text{ kg}^{-1}$ pressure drop coefficient, where the pressure drop ratio is modeled as: $\frac{dp}{dw} = -\frac{\alpha}{2p} \left(\frac{F_T}{F_{T0}} \right) \left(\frac{T}{T_0} \right)$

isothermal: $\left(\frac{T}{T_0} \right) = 1$

PBR



Limiting Reactant = B



$$P = \frac{P_0}{P}$$

$$\frac{dx}{dw} = \frac{-r'_A(x)}{F_{A0}}$$

$$\frac{dp}{dw} = -\frac{0.02}{2p} \frac{1}{1+\epsilon x} P$$

Inlet

0.7 mol/s A
0.7 mol/s B
1.4 mol/s I

$$F_I = 2.8 \frac{\text{mol}}{\text{s}}$$

$$\begin{aligned} \epsilon &= 2 - 2 - 1 \\ \epsilon &= 1 - 1 - \frac{1}{2} \\ \epsilon &= -0.5 \\ \epsilon &= -0.98 \end{aligned}$$

$$\frac{dp}{dw} = -\frac{0.02}{2p} \frac{1}{1-0.98x} P \quad (1)$$

$$P_A = P_{A0} + (-0.98)(P_{B0})(x) \frac{1}{1+\epsilon x} P$$

$$P_B = P_{B0} + (-0.98)(P_{B0})(x) \frac{1}{1+\epsilon x} P$$

$$\frac{dx}{dw} = -k \left(P_{A0} + (-0.98)(P_{B0})(x) \frac{1}{1+(-0.98)x} P \right)^{1/3} \left(P_{B0} + (-0.98)(P_{B0})(x) \frac{1}{1+(-0.98)x} P \right)^{2/3}$$

0.7 mol/s

// (2)